

SOFTWARE ENGINEERING

ASSIGNMENT 01

PROJECT REPORT

AI-Based Scam and Phishing Detection Chatbot

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2. Problem Statement

Nowadays, people receive suspicious messages, emails, and URLs that may be used for scams or phishing. Many users find it difficult to identify whether such content is safe or potentially harmful. Clicking on suspicious links or sharing personal information may lead to financial loss and privacy problems.

There is a need for a simple and user-friendly system that can help users check suspicious content, understand possible warning signs, and make safer decisions. The proposed AI-Based Scam and Phishing Detection Chatbot will analyze user-submitted messages, emails, and URLs, estimate their risk level, explain possible warning signs, and provide safety recommendations.

3. Purpose of the System

The purpose of this system is to help users identify potentially suspicious messages, emails, and links. It aims to improve awareness of online scams and phishing by providing risk estimates, explanations, and useful safety tips.

4. Users of the System

The main users of this system are:

1. **Students:** To check suspicious messages and links they receive online.
2. **General Internet Users:** To understand possible scam and phishing warnings.
3. **Employees:** To review suspicious emails and messages before interacting with them.
4. **Small Business Owners:** To help identify potentially fraudulent messages related to their business.

5. Stakeholders

The stakeholders of the system include:

1. **End Users:** People who use the chatbot to check suspicious content.
2. **Software Developers:** People responsible for designing, developing, and maintaining the system.
3. **System Administrator:** Responsible for managing the system and its technical operation.
4. **Educational Institutions:** May use the system to promote cybersecurity awareness.
5. **Cybersecurity Experts:** May provide guidance on identifying common scam and phishing indicators.

6. Functional Requirements

Functional requirements describe what the system should do.

FR1: User Input

The system shall allow users to enter suspicious messages, emails, or URLs for analysis.

FR2: Content Analysis

The system shall analyze the submitted content using predefined rules and/or an AI-based classification model.

FR3: Risk Estimation

The system shall display an estimated risk level, such as Low Risk, Medium Risk, or High Risk.

FR4: Explanation of Warning Signs

The system shall explain possible warning signs found in the submitted content, such as suspicious links, urgent language, or requests for sensitive information.

FR5: Safety Recommendations

The system shall provide safety tips and recommendations based on the analysis results.

7. Non-Functional Requirements

Non-functional requirements describe how well the system should perform.

NFR1: Usability

The system shall have a simple and user-friendly interface that is easy to understand.

NFR2: Performance

The system should analyze submitted content and display results within a reasonable response time.

NFR3: Security

The system shall protect user-submitted information and prevent unauthorized access to stored data.

NFR4: Reliability

The system should provide consistent results for the same input under the same conditions.

NFR5: Maintainability

The system should be designed so that developers can update its rules, AI model, and features easily.

8. Risks of the System

Risk 1: Incorrect Classification

The system may incorrectly classify a safe message as suspicious or fail to identify a harmful message.

Risk 2: Privacy and Data Security

User-submitted messages, emails, or URLs may contain sensitive information that could be exposed if data is not handled securely.

Risk 3: Over-Reliance on AI

Users may trust the chatbot's result completely and assume that content marked as low risk is always safe.

9. Quality Attributes

1. Usability

The system should be easy to use and provide clear explanations that users can understand.

2. Security

The system should protect user information and minimize the collection and storage of sensitive data.

3. Reliability

The system should provide stable performance and communicate uncertainty when it cannot confidently classify submitted content.

10. Where AI Could Be Used

AI could be used in the following areas:

1. **Text Classification:** To analyze messages and emails and identify patterns associated with scams or phishing.
2. **URL Analysis:** To examine URL-related features and identify potentially suspicious links.
3. **Natural Language Processing (NLP):** To understand suspicious wording, urgency, and requests for personal information.
4. **Risk Estimation:** To combine identified indicators and estimate the risk level of submitted content.
5. **Explanation Generation:** To provide understandable explanations and safety suggestions based on detected warning signs.

11. What Could Go Wrong If AI Is Used Incorrectly?

1. **False Positives:** The AI may mark a legitimate message as suspicious, causing unnecessary concern.
2. **False Negatives:** The AI may fail to detect a scam or phishing attempt, giving users a false sense of safety.

3. **Privacy Issues:** Sensitive user information may be exposed if it is collected or stored improperly.
4. **Misleading Explanations:** The chatbot may provide incorrect or confusing reasons for its classification.
5. **Overconfidence:** Users may rely entirely on AI results instead of checking suspicious content carefully.

12. Conclusion

The AI-Based Scam and Phishing Detection Chatbot is a proposed software system designed to help users recognize potentially suspicious online content. By combining content analysis, risk estimation, explanations, and safety recommendations, the system aims to improve cybersecurity awareness. However, its results should be treated as estimates rather than guaranteed proof that content is safe or fraudulent.